

Frank — AMC Upper Primary **BALANCED PRIZE TAIL Q26–Q30**

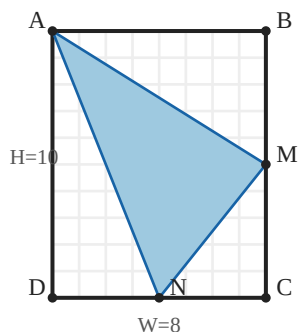
30 Jun 2026 | Five tail questions, **five different skills** — counting, residue, a repeating-map cycle, an area dissection, and a cube invariant. No two are solved the same way. Whole-number answers; show your working.

Q26. [M2] [6 marks] What is the smallest three-digit number with all of these properties: divisible by 6; contains the digit 3; has no digit 0; has all different digits; has exactly 1 odd digit?

Q27. [M5] [7 marks] In how many ways can you choose 3 different numbers from 2 to 22 so that their sum is a multiple of 3?

Q28. [M6] [8 marks] Start with the number 8665. At each step replace the number with the sum of the squares of its digits (for example $24 \rightarrow 2^2 + 4^2 = 20$). What is the 3996th number in this list?

Q29. [M7] [9 marks] ABCD is a rectangle measuring 8 by 10 (area 80). M is the midpoint of side BC and N is the midpoint of side DC. M and N are each joined to A and to each other. What is the area of the shaded triangle AMN?



Q30. [M8] [10 marks] Using only the digits 1, 2, 4, each face of a cube shows a different 3-digit number. The three faces meeting at one corner show 214, 222 and 224. The two numbers on each pair of opposite faces add to the same total. What is the largest that this total could be?

Each item is a code-verified twin of a real AMC anchor (unique answer, difficulty in band). The geometry figure is drawn from the SAME exact coordinates used to compute its area, so the picture and the answer cannot disagree.